

Princeton University  
Mathematical Intermediate Microeconomics (ECO 310)  
Midterm Spring 2022 Solutions

**Instructions:** You have 80 minutes to complete this exam, plus 10 minutes to upload your answers. Each part of each question (Q1, Q2, and Q3) is worth 10, for a total of 100. **Show your work and explain your answers for full credit!**

- The exam is open-book, and calculators are permitted. However, please work **alone**.
- **If you are unsure whether you're interpreting a question correctly, state your interpretation and proceed (or email me if you prefer).**

The honor code applies, as usual.

## Q1: Proofs and General Concepts

- a. Sally has preferences over  $X = \mathbb{R}^2$  represented by  $u(x_1, x_2) = \max\{x_1, x_2\}$ . Are her preferences (i) rational? (ii) convex?
- b. Now Sally tells you that she has a utility function  $u : X \rightarrow \mathbb{R}$ , and, from any menu  $A \subseteq X$ , chooses the option in  $A$  with the lowest utility ( $C(A) = \arg \min_{x \in A} u(x)$ ). Do her choices satisfy IIA? (proof or counterexample)
- c. Using only the assumption that utility of expected wealth beats expected utility, prove that a gamble with 50-50 odds of winning or losing \$100 is preferred to a gamble with 50-50 odds of winning or losing \$200.

## Q2: Demand

Homer Simpson's preferences over beer ( $B$ ) and leisure ( $L$ ) are given by

$$u(B, L) = \min\{B, L\}$$

He spends 8h per day passed out, but can work as many of the remaining 16 hours as he wishes, at a flat wage  $w$ /hour. Marge also gives him an allowance of  $N$  per day. He buys cheap beer, with price  $p_B = 1$ .

- Find his optimal labor supply,  $\ell(w)$ . (i.e., how many hours will he optimally work at wage  $w$ ?)
- Calculate his compensating variation if the price of beer doubles.
- How much allowance ( $N$ ) does Homer need at beer price  $p_B$  and wage rate  $w$  to get utility  $u^*$ ? (Assume  $u^* \leq 16$ )

## Q3: Expected Utility

Sam has 60 minutes to complete his midterm, which has both an easy question and a hard question. For parts (a)-(c), assume that he can choose how to split his time between the two questions. Every minute spent on the easy question earns a guaranteed point. Points on the hard question are uncertain, as Sam is not sure whether he'll figure it out: he expects that if he spends  $x$  minutes on it, he'll get  $2x$  points with probability  $p$ , and zero points otherwise.

- Write out Sam's expected utility if he spends  $x$  minutes on the hard question and the remaining  $60 - x$  minutes on the easy question, assuming utility  $\sqrt{T}$  from point total  $T$ .
- Suppose (for this part only) that  $p < \frac{1}{2}$ . Show that in expectation, each minute spent on the hard question earns less than a point, and explain why this implies that Sam should devote the entire 60 minutes to the easy question.
- Assuming now that  $p > \frac{1}{2}$ , find the optimal # minutes  $x$  that Sam should spend on the hard question. (Partial minutes **are** allowed).
- Now modify the question slightly. Suppose Sam has to choose at the outset to do the hard question **OR** the easy question, he cannot attempt both. For what values of  $p$  will he choose the hard question?